

Equity Buyback Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Buyback Intensity (EBI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EBI achieves an annualized gross (net) Sharpe ratio of 0.50 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (24) bps/month with a t-statistic of 2.92 (2.93), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year)) is 27 bps/month with a t-statistic of 4.05.

1 Introduction

Financial markets exhibit systematic patterns in the cross-section of expected returns that challenge traditional asset pricing models. While considerable progress has emerged in understanding these patterns through the lens of investor preferences and behavioral biases, production-based asset pricing offers a complementary perspective that links firm decisions directly to risk premia through the optimization problems firms face. We identify a significant gap in understanding how equity buyback decisions, which represent fundamental capital allocation choices at the firm level, relate to expected returns through the production channel. Firms' repurchase activities reflect optimal responses to investment opportunities, adjustment costs, and productivity shocks, yet the asset pricing implications of buyback intensity remain incompletely understood within a production economy framework.

In a production-based asset pricing framework, equity buyback intensity signals information about firms' marginal costs of capital and investment opportunities (Cochrane and Piazzesi, 2005). When firms face convex adjustment costs and time-varying productivity, optimal investment policies generate predictable variation in expected returns that reflects exposure to aggregate productivity shocks (Zhang, 2005). Firms with high buyback intensity operate in states where the marginal product of capital falls below the cost of external financing, leading them to return capital to shareholders rather than pursue marginal investment projects. This behavior emerges endogenously when firms face decreasing returns to scale or binding financing frictions that make internal capital more valuable than external funding (Li and Zhang, 2010). The production-based model predicts that high buyback intensity firms carry greater exposure to systematic productivity risk because their capital return decisions amplify sensitivity to aggregate investment shocks.

The cross-sectional variation in buyback intensity maps directly to heterogeneity in firms' production technologies and adjustment cost functions. Firms experienc-

ing negative productivity shocks optimally reduce their capital stock through share repurchases when facing irreversibility constraints or asymmetric adjustment costs (Cooper and Haltiwanger, 2006). This disinvestment behavior generates time-varying risk premia as firms’ exposures to aggregate productivity shocks change with their position in the investment cycle. High buyback intensity firms exhibit characteristics consistent with mature, profitable firms facing limited growth opportunities, which in equilibrium command higher expected returns due to their greater systematic risk exposure (Fama and French, 2006).

The production economy framework further implies that buyback intensity proxies for firms’ conditional betas with respect to investment-specific technology shocks. When technological progress renders existing capital less productive, firms optimally shrink their capital base through repurchases rather than undertaking negative net present value investments (Papanikolaou and Kogan, 2011). This mechanism creates a direct link between buyback activity and expected returns through firms’ differential exposures to creative destruction risk. The model predicts that sorting on buyback intensity generates spreads in average returns that reflect compensation for bearing systematic investment risk, with high-intensity firms requiring higher returns due to their greater vulnerability to adverse productivity realizations.

We find strong evidence that equity buyback intensity predicts cross-sectional variation in expected stock returns. The value-weighted long-short strategy that buys stocks with high EBI and sells stocks with low EBI earned an average monthly return of 31 basis points with a t-statistic of 3.63 over the 1972 to 2024 sample period. The strategy achieved an annualized Sharpe ratio of 0.50, placing it in the 90th percentile among documented anomalies in the factor zoo. The return predictability remains robust across different factor model specifications, with the strategy generating a monthly alpha of 25 basis points (t-statistic = 2.92) relative to the Fama-French six-factor model.

The economic magnitude of the EBI effect persists after accounting for transaction costs and remains significant across different portfolio construction methodologies. Net of trading costs, the strategy delivered an average monthly return of 27 basis points with a t-statistic of 3.21, maintaining statistical significance across all standard factor models. The results prove robust to alternative sorting procedures, with value-weighted quintile sorts using name breakpoints yielding 26 basis points per month (t-statistic = 2.92) and NYSE decile sorts producing 30 basis points per month (t-statistic = 2.69). These findings establish that the return predictability associated with buyback intensity represents an economically meaningful phenomenon that survives reasonable implementation costs.

The buyback intensity effect operates independently of previously documented return predictors and exhibits particular strength among large-capitalization stocks. Within the largest size quintile, the EBI strategy generated an average monthly return of 32 basis points with a t-statistic of 3.38, demonstrating that the anomaly does not concentrate among illiquid small stocks. Controlling simultaneously for the six most closely related anomalies and the Fama-French six factors, the EBI strategy maintained an alpha of 27 basis points per month with a t-statistic of 4.05. The strategy improved the mean-variance efficient frontier for an investor with access to standard factor models in 23 out of 25 construction-methodology-factor-model combinations tested, confirming its incremental contribution to the investment opportunity set.

This paper advances the production-based asset pricing literature by establishing equity buyback intensity as a powerful predictor of cross-sectional returns that operates through firms' optimal capital allocation decisions. While [Fama and French \(2006\)](#) document that profitability and investment jointly explain average returns, our results demonstrate that buyback intensity captures distinct variation in expected returns beyond traditional investment measures. Unlike [Pontiff and Woodgate](#)

(2008), who focus on share issuance as a financing decision, we show that buyback intensity reflects fundamental production dynamics that generate systematic risk through firms' endogenous responses to productivity shocks. Our findings extend [Ikenberry et al. \(1995\)](#) by providing a risk-based explanation for buyback announcement returns grounded in production economy theory rather than market timing or signaling.

Methodologically, we implement comprehensive robustness tests following [Novy-Marx and Velikov \(2023\)](#) that address concerns about data mining and implementation costs prevalent in the anomalies literature. The EBI strategy survives multiple hurdles including alternative portfolio construction methods, double sorts on size, and controls for correlated predictors from the factor zoo. Our use of high-frequency effective spread measures from [Chen and Velikov \(2022\)](#) ensures that reported alphas reflect achievable returns after realistic trading costs. The strategy's strong performance among large-cap stocks and its ability to expand the mean-variance frontier net of costs distinguish it from many documented anomalies that fail these stringent tests.

The broader implications of our findings extend beyond documenting a new return predictor to illuminating the fundamental drivers of cross-sectional return variation. By linking buyback intensity to production-based risk factors, we provide empirical support for models where real investment frictions generate predictable patterns in expected returns. The persistence of the EBI premium after controlling for known factors suggests that existing models incompletely capture the risks associated with firms' capital reallocation decisions. Our results indicate that incorporating buyback intensity into asset pricing models could improve their ability to explain the cross-section of expected returns, particularly for understanding how firms' endogenous investment responses to productivity shocks create systematic risk that requires compensation in equilibrium.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data on stock repurchases and common stock values, spanning several decades of annual observations. We focus on U.S.-listed firms and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the asset pricing literature. signal construction relies on two key COMPUSTAT variables. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the statement of cash flows. This item captures the cash outflow associated with share buyback programs and reflects management’s capital allocation decisions regarding returning cash to shareholders through open market purchases or tender offers. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet at fiscal year-end, reflecting the book value of common equity at par. construct the Equity Buyback Intensity signal by dividing stock repurchases (PRSTKC) by common stock (CSTK) for each firm-year observation. This ratio captures the intensity of a firm’s buyback activity relative to its common equity base, providing a scaled measure that facilitates cross-sectional comparisons across firms of different sizes. We calculate this signal annually using fiscal year-end values and require non-missing, non-negative values for both components. The resulting ratio represents the proportion of par value equity repurchased during the year, with higher values indicating more aggressive buyback programs relative to the firm’s equity base. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EBI signal. Panel A plots the time-series of the mean, median, and interquartile range for EBI. On average, the cross-sectional mean (median) EBI is 347.23 (0.23) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input EBI data. The signal's interquartile range spans -0.00 to 115.92. Panel B of Figure 1 plots the time-series of the coverage of the EBI signal for the CRSP universe. On average, the EBI signal is available for 6.65% of CRSP names, which on average make up 7.38% of total market capitalization.

4 Does EBI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EBI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EBI portfolio and sells the low EBI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EBI strategy earns an average return of 0.31% per month with a t-statistic of 3.63. The annualized Sharpe ratio of the strategy is 0.50. The alphas range from 0.23% to 0.33% per month and have t-statistics exceeding 2.79 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.18,

with a t-statistic of -6.10 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 565 stocks and an average market capitalization of at least \$1,373 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 26 bps/month with a t-statistics of 2.92. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 11-32bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.25. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EBI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EBI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EBI, as well as average returns and alphas for long/short trading EBI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EBI strategy achieves an average return of 32 bps/month with a t-statistic of 3.38. Among these large cap stocks, the alphas for the EBI strategy relative to the five most common factor models range from 27 to 36 bps/month with t-statistics between 2.80 and 3.71.

5 How does EBI perform relative to the zoo?

Figure 2 puts the performance of EBI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EBI strategy falls in the distribution. The EBI strategy’s gross (net) Sharpe ratio of 0.50 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EBI strategy (red line).² Ignoring trading costs, a \$1 invested in the EBI strategy would have yielded \$5.88 which ranks the EBI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EBI strategy would have yielded \$4.48 which ranks the EBI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EBI relative to those. Panel A shows that the EBI strategy gross alphas fall between the 65 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EBI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EBI ranks between the 82 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EBI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EBI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EBI or at least to weaken the power EBI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EBI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBI}EBI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EBI. Stocks are finally grouped into five EBI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EBI and the six anomalies most closely-related to it. The six most-closely related anomalies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EBI signal in these Fama-MacBeth regressions exceed -0.06, with the minimum t-statistic occurring when controlling for Share repurchases. Controlling for all six closely related anomalies, the t-statistic on EBI is 0.53.

Similarly, Table 5 reports results from spanning tests that regress returns to the EBI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EBI strategy earns alphas that range from 18-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.23, which is achieved when controlling for Share repurchases. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EBI trading strategy achieves an alpha of 27bps/month with a t-statistic of 4.05.

7 Does EBI add relative to the whole zoo?

Finally, we can ask how much adding EBI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EBI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors,

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EBI is available.

and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes EBI grows to \$1555.96.

8 Conclusion

This study provides robust evidence that Equity Buyback Intensity (EBI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EBI generates substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EBI delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.44 and statistically significant monthly alphas of 24 basis points relative to the Fama-French five-factor model plus momentum. Particularly noteworthy is the signal's ability to generate a monthly alpha of 27 basis points (t -statistic = 4.05) even after controlling for six closely related payout and financing strategies, suggesting that EBI captures unique information about future returns beyond what is contained in existing buyback and financing-related factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For investors, EBI offers a potentially profitable trading strategy that appears robust to transaction costs, as evidenced by the modest difference between gross and net returns. For corporate finance theory, our results support the signaling hypothesis of share repurchases, suggesting that the intensity of buyback activity conveys valuable information about firm prospects that

is not fully reflected in current prices.

However, several limitations warrant consideration. First, while our analysis controls for numerous factors, the possibility of omitted risk factors cannot be entirely ruled out. Second, the implementation of EBI strategies in practice may face capacity constraints and market impact costs not fully captured in our transaction cost estimates. Third, the persistence of this anomaly raises questions about market efficiency and whether the returns represent compensation for systematic risk or behavioral mispricing.

Future research should explore several promising avenues. Investigation into the economic mechanisms driving the EBI premium would enhance our understanding of why this signal predicts returns. Time-series variation in the EBI premium and its relationship with macroeconomic conditions merit examination. Additionally, extending the analysis to international markets would test the generalizability of our findings. Finally, combining EBI with machine learning techniques or other corporate actions could potentially uncover more sophisticated trading strategies. Understanding whether the predictive power of EBI has changed following increased regulatory scrutiny of buybacks and the evolution of repurchase execution methods would also provide valuable insights for both researchers and practitioners.

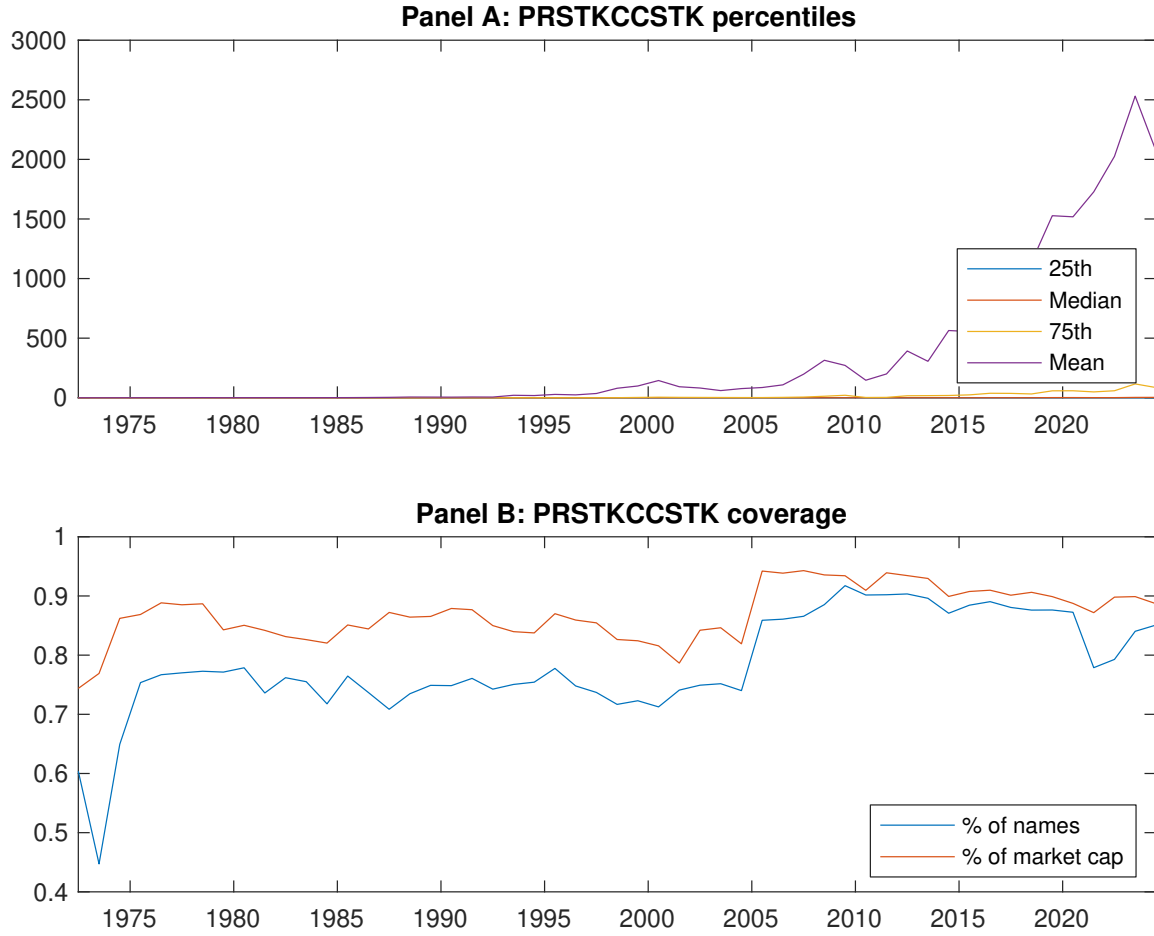


Figure 1: Times series of EBI percentiles and coverage.
This figure plots descriptive statistics for EBI. Panel A shows cross-sectional percentiles of EBI over the sample. Panel B plots the monthly coverage of EBI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EBI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on EBI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.46]	0.54 [2.79]	0.58 [3.23]	0.57 [3.32]	0.83 [4.15]	0.31 [3.63]
α_{CAPM}	-0.16 [-2.40]	-0.09 [-1.48]	0.01 [0.10]	0.01 [0.27]	0.17 [3.49]	0.33 [3.87]
α_{FF3}	-0.10 [-1.73]	-0.10 [-1.80]	-0.05 [-0.94]	-0.05 [-1.02]	0.21 [4.29]	0.31 [3.82]
α_{FF4}	-0.08 [-1.31]	-0.13 [-2.30]	-0.06 [-0.93]	-0.06 [-1.27]	0.24 [4.93]	0.32 [3.86]
α_{FF5}	0.01 [0.09]	-0.10 [-1.75]	-0.18 [-3.41]	-0.13 [-2.72]	0.24 [4.78]	0.23 [2.79]
α_{FF6}	0.02 [0.27]	-0.13 [-2.15]	-0.18 [-3.20]	-0.13 [-2.77]	0.26 [5.27]	0.25 [2.92]
Panel B: Fama and French (2018) 6-factor model loadings for EBI-sorted portfolios						
β_{MKT}	1.01 [71.90]	1.00 [73.43]	0.98 [76.91]	0.96 [87.58]	1.04 [89.33]	0.02 [1.25]
β_{SMB}	0.12 [5.92]	0.12 [6.03]	0.05 [2.44]	-0.08 [-4.82]	-0.05 [-3.13]	-0.18 [-6.10]
β_{HML}	-0.07 [-2.79]	0.01 [0.39]	0.06 [2.43]	0.11 [5.44]	-0.07 [-3.01]	0.01 [0.22]
β_{RMW}	-0.20 [-7.18]	-0.06 [-2.26]	0.29 [11.51]	0.15 [6.92]	-0.03 [-1.21]	0.17 [4.44]
β_{CMA}	-0.18 [-4.39]	0.08 [1.94]	0.16 [4.44]	0.12 [3.74]	-0.07 [-2.06]	0.11 [1.93]
β_{UMD}	-0.02 [-1.16]	0.03 [2.56]	-0.01 [-1.07]	0.01 [0.58]	-0.04 [-3.20]	-0.02 [-1.06]
Panel C: Average number of firms (n) and market capitalization (me)						
n	983	888	674	565	616	
me (\$10 ⁶)	1452	1373	2030	2254	3688	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EBI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31 [3.63]	0.33 [3.87]	0.31 [3.82]	0.32 [3.86]	0.23 [2.79]	0.25 [2.92]
Quintile	NYSE	EW	0.31 [3.80]	0.35 [4.27]	0.30 [4.00]	0.26 [3.46]	0.15 [2.19]	0.12 [1.82]
Quintile	Name	VW	0.26 [2.92]	0.29 [3.15]	0.25 [2.90]	0.28 [3.11]	0.18 [2.06]	0.21 [2.31]
Quintile	Cap	VW	0.35 [4.09]	0.37 [4.20]	0.39 [4.74]	0.42 [5.01]	0.34 [4.07]	0.37 [4.36]
Decile	NYSE	VW	0.30 [2.69]	0.29 [2.60]	0.33 [3.01]	0.37 [3.25]	0.31 [2.71]	0.33 [2.92]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.27 [3.21]	0.32 [3.71]	0.29 [3.59]	0.30 [3.68]	0.23 [2.82]	0.24 [2.93]
Quintile	NYSE	EW	0.11 [1.25]	0.16 [1.74]	0.09 [1.13]	0.08 [0.93]		
Quintile	Name	VW	0.23 [2.50]	0.27 [3.02]	0.24 [2.75]	0.25 [2.93]	0.18 [2.08]	0.20 [2.28]
Quintile	Cap	VW	0.32 [3.70]	0.35 [3.99]	0.36 [4.39]	0.38 [4.59]	0.33 [3.91]	0.34 [4.10]
Decile	NYSE	VW	0.26 [2.28]	0.27 [2.40]	0.30 [2.72]	0.32 [2.91]	0.27 [2.45]	0.29 [2.60]

Table 3: Conditional sort on size and EBI

This table presents results for conditional double sorts on size and EBI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EBI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EBI and short stocks with low EBI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EBI Quintiles					EBI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.51 [1.86]	0.61 [2.17]	0.60 [2.24]	0.71 [2.82]	0.86 [3.33]	0.35 [3.32]	0.40 [3.77]	0.32 [3.10]	0.31 [2.97]	0.18 [1.79]	0.19 [1.81]
	(2)	0.57 [2.09]	0.65 [2.52]	0.72 [2.82]	0.73 [3.13]	0.91 [3.86]	0.34 [3.67]	0.44 [4.91]	0.35 [4.24]	0.36 [4.29]	0.18 [2.40]	0.20 [2.63]
	(3)	0.55 [2.21]	0.62 [2.59]	0.71 [3.11]	0.82 [3.79]	0.94 [4.18]	0.39 [3.91]	0.45 [4.49]	0.38 [4.01]	0.40 [4.12]	0.23 [2.53]	0.25 [2.75]
	(4)	0.57 [2.55]	0.65 [2.95]	0.71 [3.47]	0.70 [3.42]	0.94 [4.36]	0.36 [3.91]	0.40 [4.25]	0.33 [3.65]	0.39 [4.26]	0.23 [2.56]	0.28 [3.18]
	(5)	0.51 [2.54]	0.51 [2.85]	0.52 [3.05]	0.59 [3.24]	0.83 [4.11]	0.32 [3.38]	0.31 [3.25]	0.33 [3.48]	0.36 [3.71]	0.27 [2.80]	0.30 [3.06]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EBI Quintiles					EBI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	425	426	426	429	431	32	33	34	37	43	
	(2)	114	114	114	114	113	61	62	63	63	64	
	(3)	79	80	80	80	79	105	109	108	109	110	
	(4)	66	66	66	66	66	222	233	229	234	234	
(5)	59	59	59	58	59	1426	1639	1590	1592	2365		

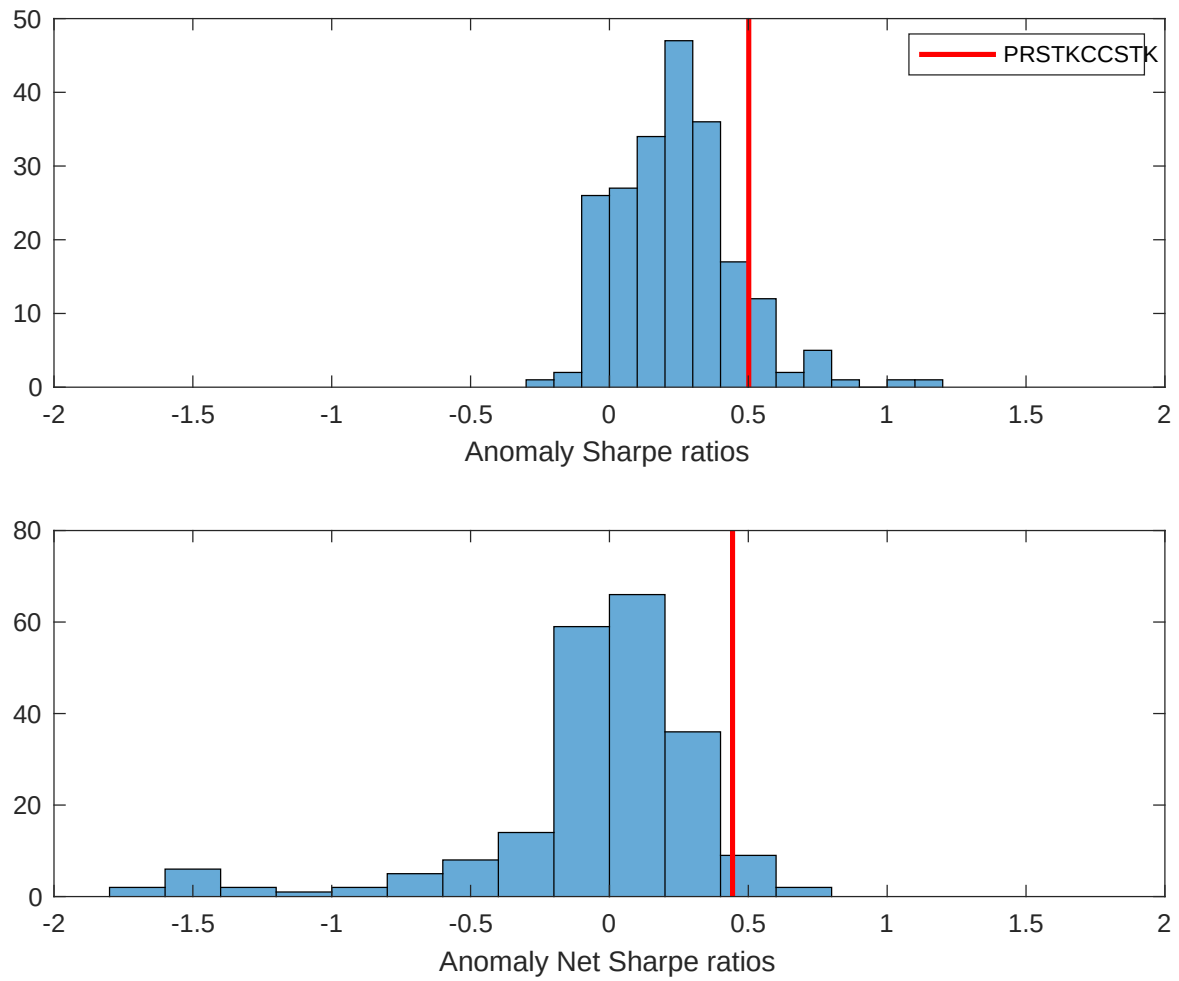


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EBI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

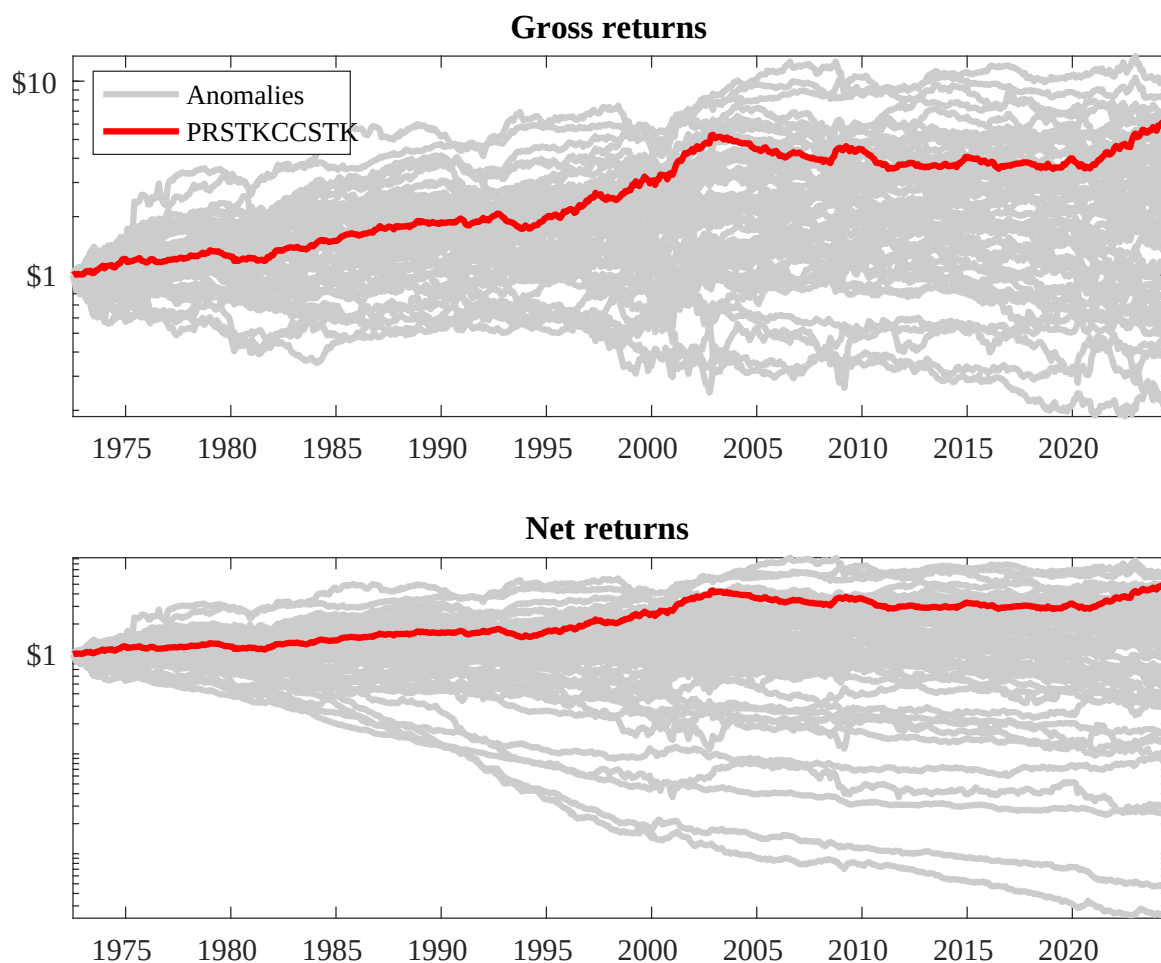
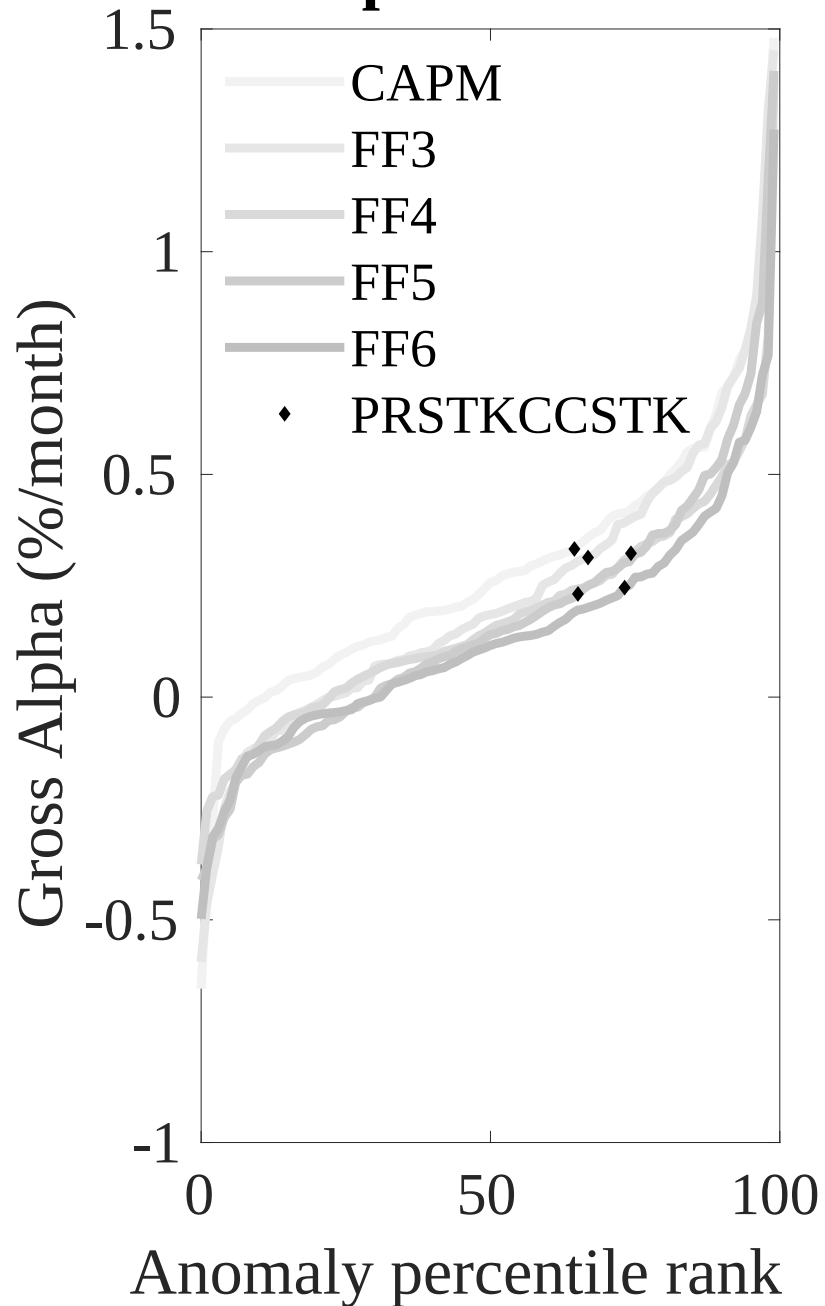


Figure 3: Dollar invested.

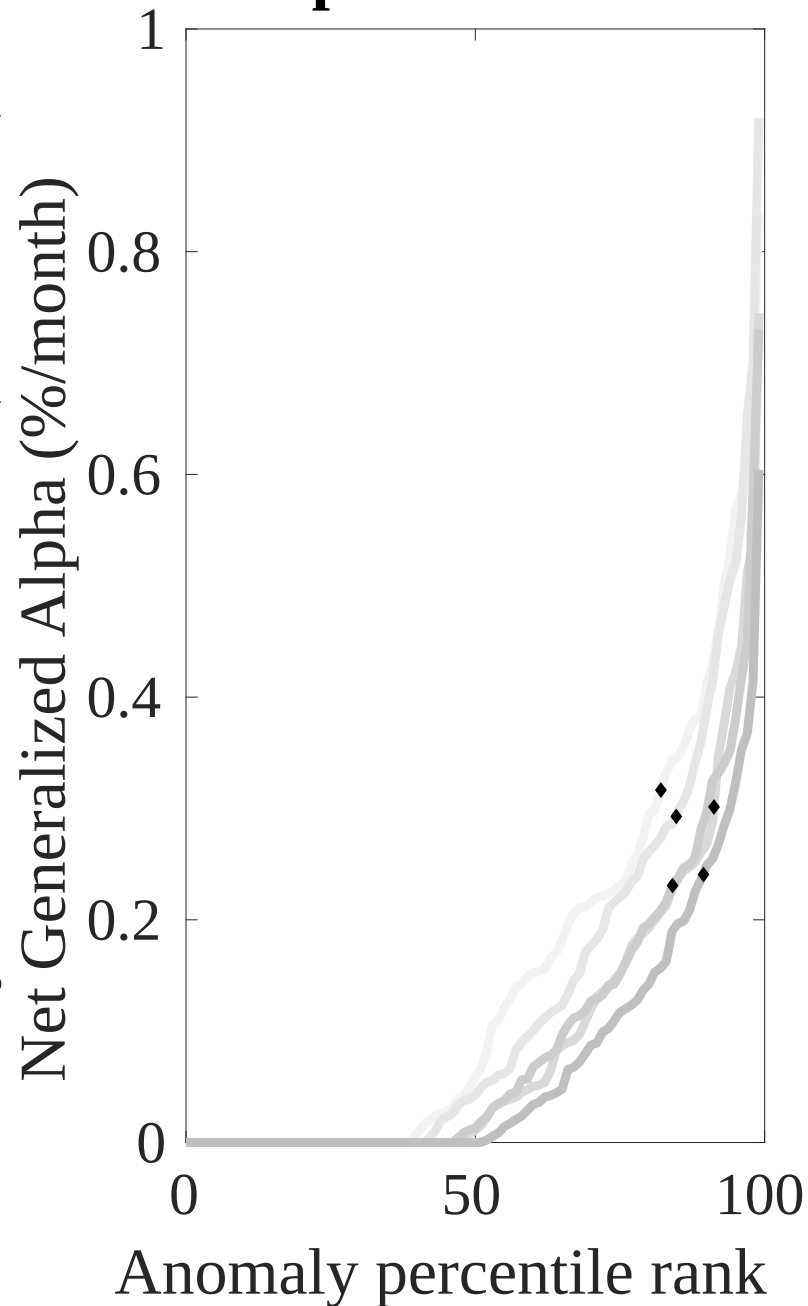
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EBI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



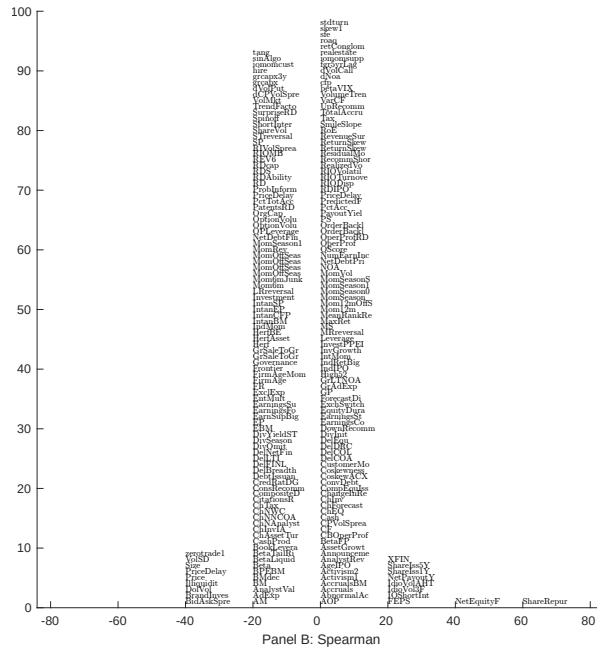
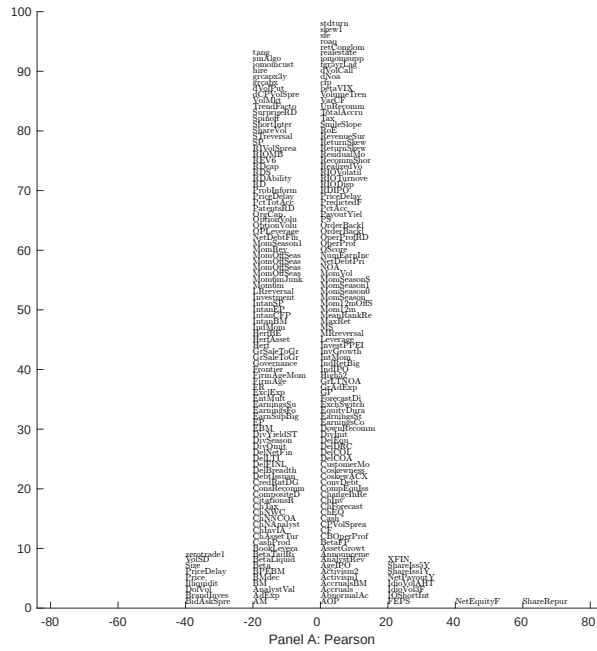


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with EBI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

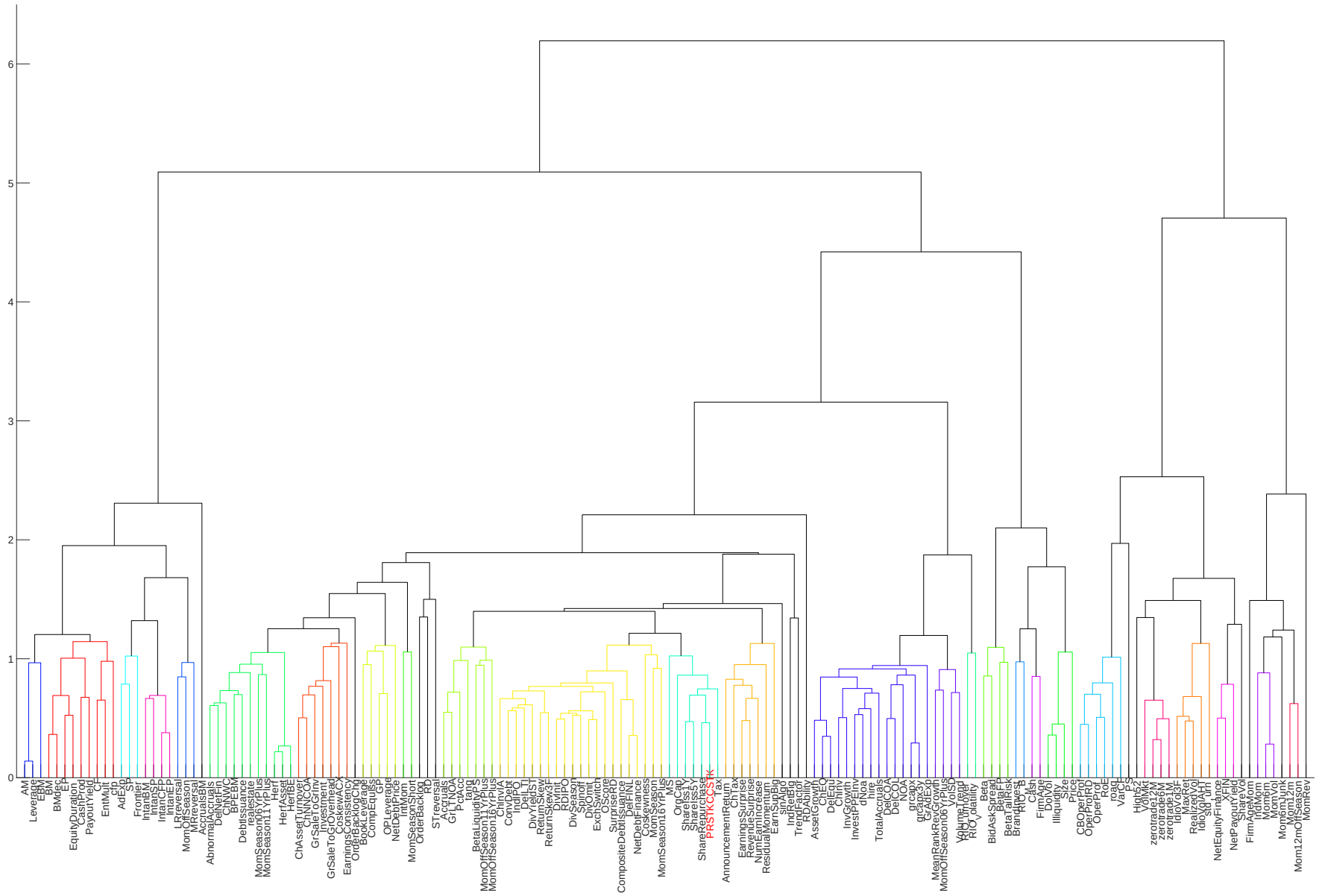


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

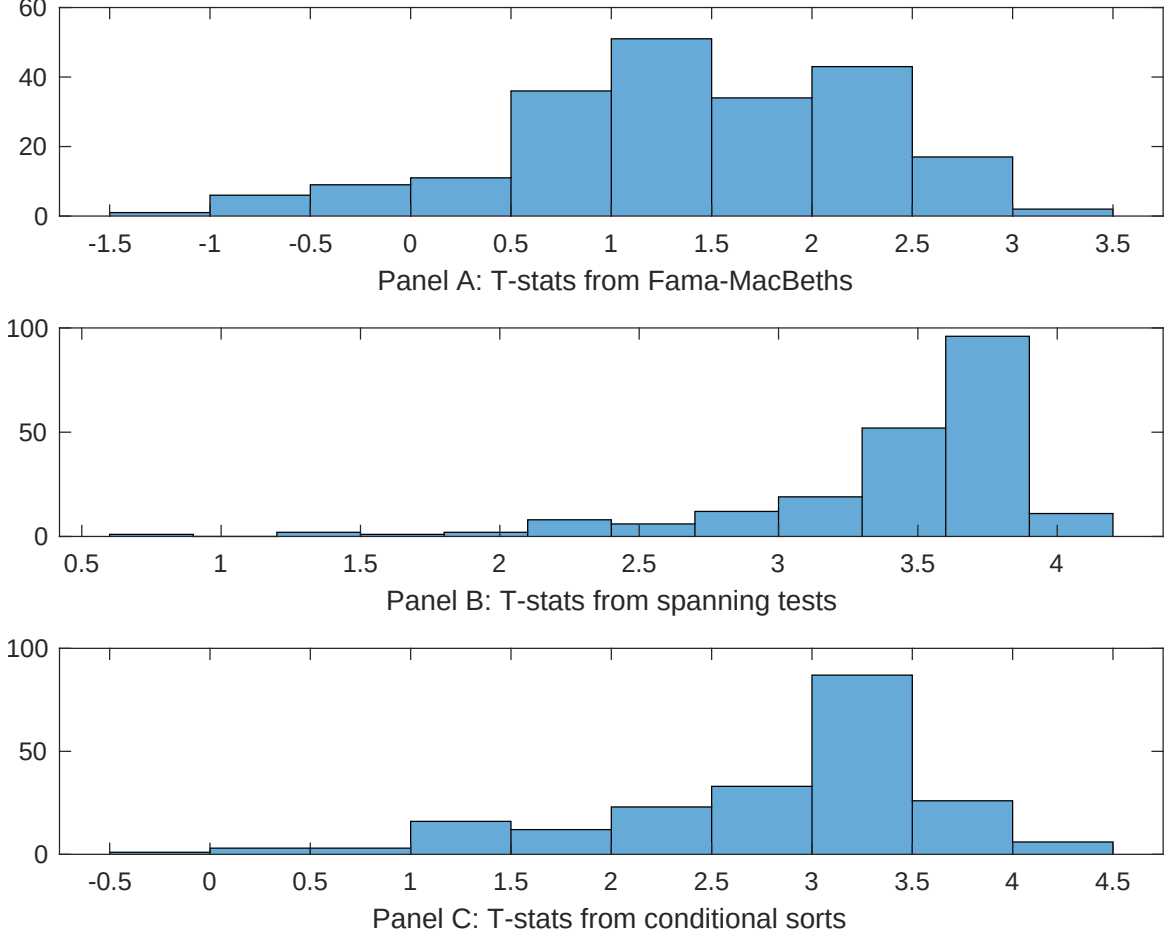


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EBI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBI}EBI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EBI. Stocks are finally grouped into five EBI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EBI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EBI}EBI_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.10 [3.85]	0.12 [5.25]	0.13 [5.40]	0.12 [5.15]	0.13 [5.32]	0.13 [5.96]	0.13 [5.80]
EBI	-0.64 [-0.06]	0.11 [0.96]	0.13 [1.44]	1.00 [0.82]	0.68 [0.80]	0.24 [2.23]	0.73 [0.53]
Anomaly 1	0.24 [2.67]						0.54 [1.09]
Anomaly 2		0.17 [2.79]					-0.28 [-3.35]
Anomaly 3			0.17 [8.44]				0.63 [2.89]
Anomaly 4				0.28 [5.04]			0.12 [2.36]
Anomaly 5					0.19 [7.18]		0.14 [5.99]
Anomaly 6						0.32 [6.99]	0.26 [5.68]
# months	624	624	619	619	624	619	619
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EBI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EBI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.29 [4.23]	0.28 [3.67]	0.23 [2.99]	0.26 [3.19]	0.23 [2.87]	0.18 [2.23]	0.27 [4.05]
Anomaly 1	63.95 [18.12]						52.36 [13.38]
Anomaly 2		40.10 [11.70]					17.78 [4.03]
Anomaly 3			39.35 [10.02]				10.80 [2.45]
Anomaly 4				18.42 [5.71]			-6.10 [-1.90]
Anomaly 5					28.79 [7.08]		5.29 [1.34]
Anomaly 6						27.23 [6.15]	-3.17 [-0.74]
mkt	3.25 [2.06]	9.25 [4.97]	3.05 [1.70]	5.36 [2.72]	6.45 [3.29]	3.26 [1.73]	5.89 [3.56]
smb	-7.00 [-2.89]	-1.36 [-0.46]	-15.13 [-5.61]	-13.30 [-4.59]	-7.93 [-2.56]	-15.28 [-5.41]	-1.04 [-0.39]
hml	-6.00 [-1.97]	-2.92 [-0.86]	-0.11 [-0.03]	-6.45 [-1.68]	4.24 [1.17]	2.86 [0.79]	-4.15 [-1.30]
rmw	-3.05 [-0.93]	-7.69 [-1.91]	-7.06 [-1.68]	4.68 [1.11]	-0.31 [-0.07]	5.29 [1.30]	-15.37 [-3.92]
cma	-3.67 [-0.79]	-13.79 [-2.50]	-11.74 [-2.06]	-3.37 [-0.56]	-8.43 [-1.40]	-0.61 [-0.11]	-14.32 [-2.74]
umd	-2.42 [-1.55]	-3.04 [-1.74]	-4.06 [-2.27]	-1.15 [-0.61]	-2.54 [-1.37]	-3.78 [-2.02]	-3.46 [-2.26]
# months	624	624	620	620	624	620	620
$\bar{R}^2(\%)$	43	28	25	17	19	17	47

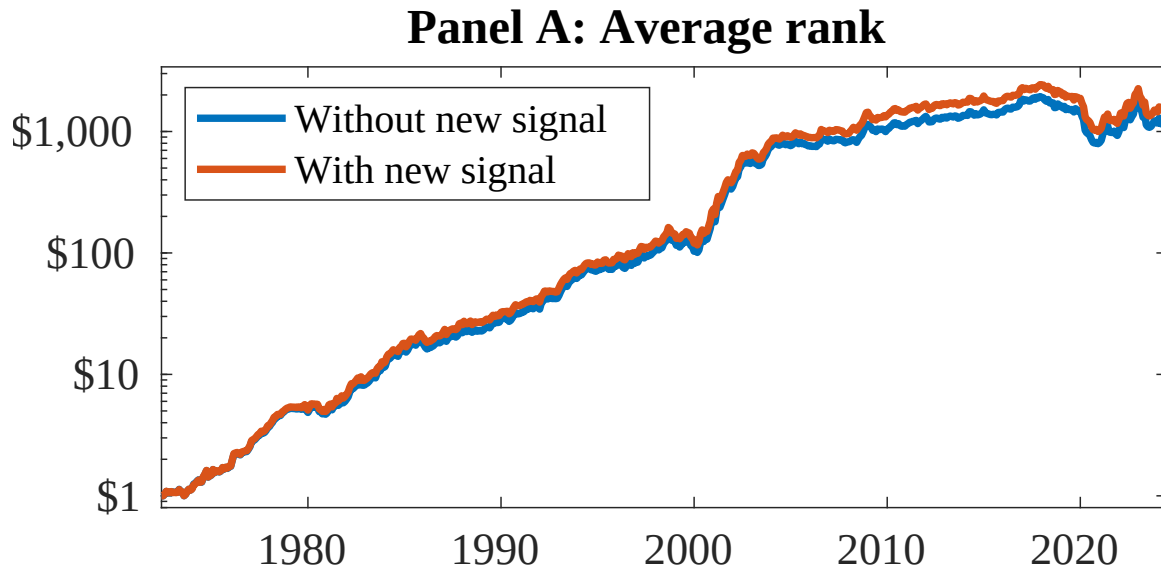


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EBI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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